

# INCIDENT RESPONSE REPORT

## Brute Force SSH Attack — Detection & Analysis

Prepared by: Rutuja Late

Date: June 27, 2026

Classification: Confidential

Severity: High

### 1. EXECUTIVE SUMMARY

On June 25, 2026, a brute force SSH attack was detected against an internal system (127.0.0.1) in a controlled lab environment. The attacker used an automated tool (Hydra with libssh\_0.11.2 library) to attempt multiple password combinations against the SSH service running on port 22.

The attack resulted in unauthorized access after 5 failed login attempts within 5 seconds. The incident was detected using Splunk SIEM through automated alert configured with threshold-based detection logic. Network-level evidence was captured and analyzed using Wireshark.

Severity: High

Status: Detected & Analyzed

MITRE ATT&CK: T1110.001 — Password Guessing

### 2. INCIDENT DETAILS

Incident ID: IR-2026-001

Date & Time: June 25, 2026 — 17:19:51 to 17:19:56 IST

Detection Time: June 27, 2026 — 10:45:01 IST (Automated Splunk Alert)

Analyst: Rutuja Late

Severity: High

Status: Closed

Affected System:

- Hostname: kali
- IP Address: 127.0.0.1
- Service: SSH (Port 22)
- OS: Kali Linux 2025.3

Attacker:

- IP Address: 127.0.0.1 (localhost — lab simulation)
- Tool Used: Hydra v9.5 (libssh\_0.11.2)

### 3. ATTACK TIMELINE

17:19:51 — Hydra initiated brute force attack against SSH port 22  
17:19:51 — Failed login attempt #1 (password: password123)  
17:19:51 — Failed login attempt #2 (password: 123456)  
17:19:51 — Failed login attempt #3 (password: admin123)  
17:19:52 — Failed login attempt #4 (password: letmein)  
17:19:52 — Failed login attempt #5 (password: qwerty)  
17:19:52 — Successful login (password: kali) — unauthorized access achieved  
17:19:52 — Session opened for user kali  
17:19:52 — Session closed  
17:19:56 — Hydra attack completed

Total attack duration: 5 seconds

Failed attempts: 5

Successful logins: 1

### 4. EVIDENCE

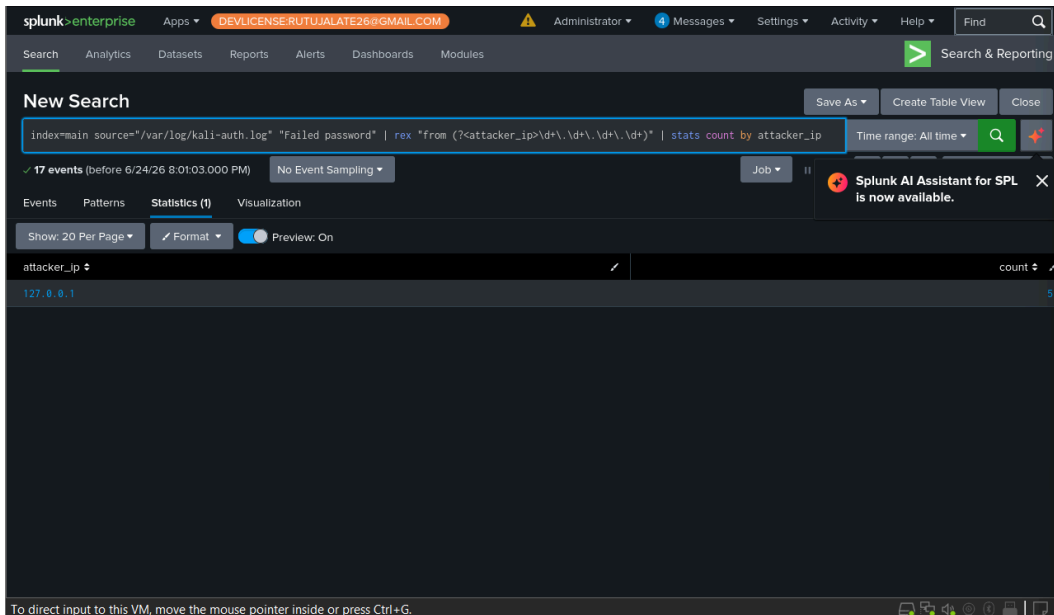
#### 4.1 Splunk SIEM Evidence

Detection Query:

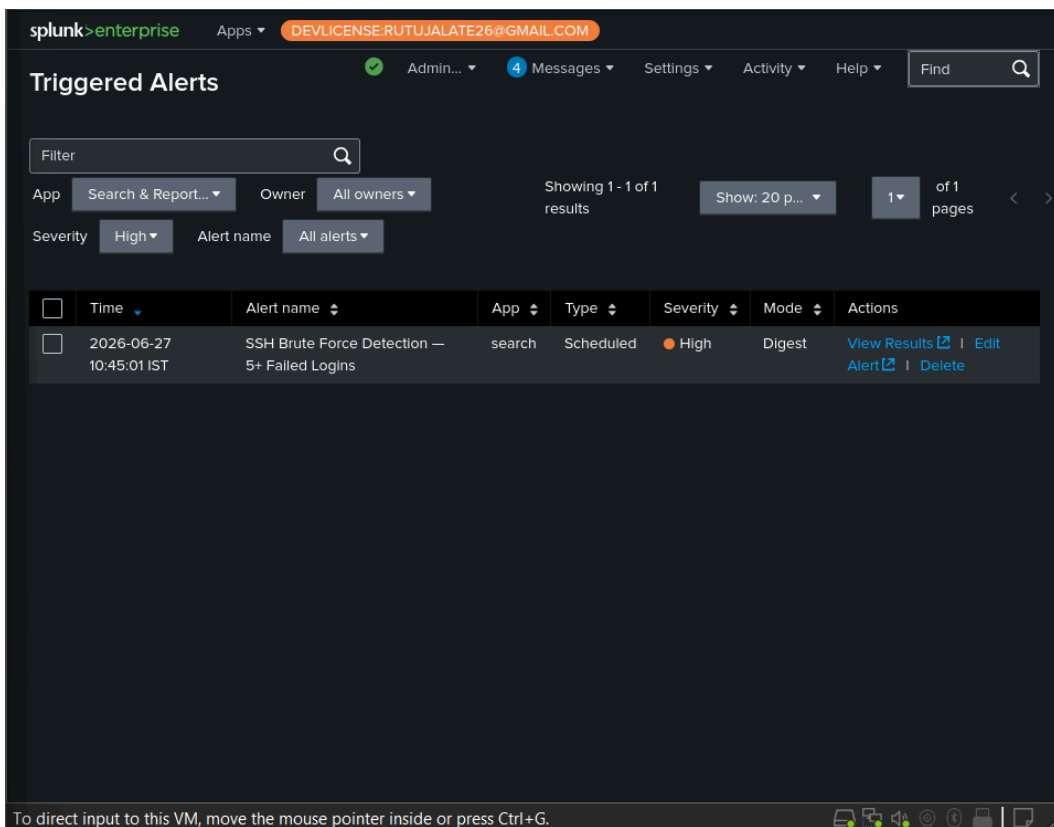
```
index=main source="/var/log/ssh-bruteforce.log" "Failed password"  
| rex "from (?<attacker_ip>\d+\.\d+\.\d+\.\d+)"  
| stats count by attacker_ip  
| where count >= 5
```

Findings:

- 5 "Failed password" events from 127.0.0.1
- 1 "Accepted password" event for user kali
- Automated alert triggered at 10:45:01 IST
- Alert Name: SSH Brute Force Detection — 5+ Failed Logins
- Alert Severity: High



**Figure 1: Splunk — Attacker IP Breakdown (127.0.0.1, Count: 5)**



**Figure 2: Splunk — Triggered Alert (High Severity)**

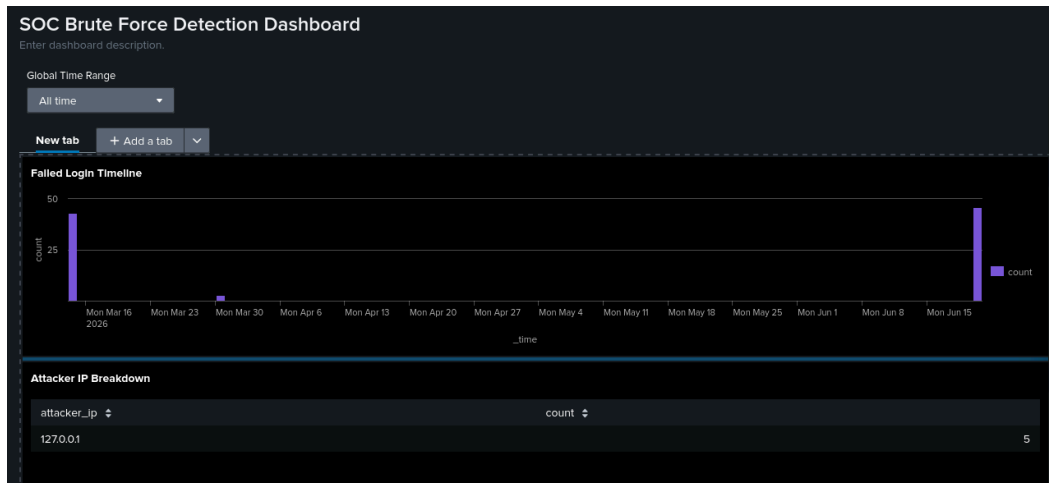


Figure 3: Splunk — SOC Dashboard

## 4.2 Wireshark Network Evidence

Capture File: brute-force-capture.pcap

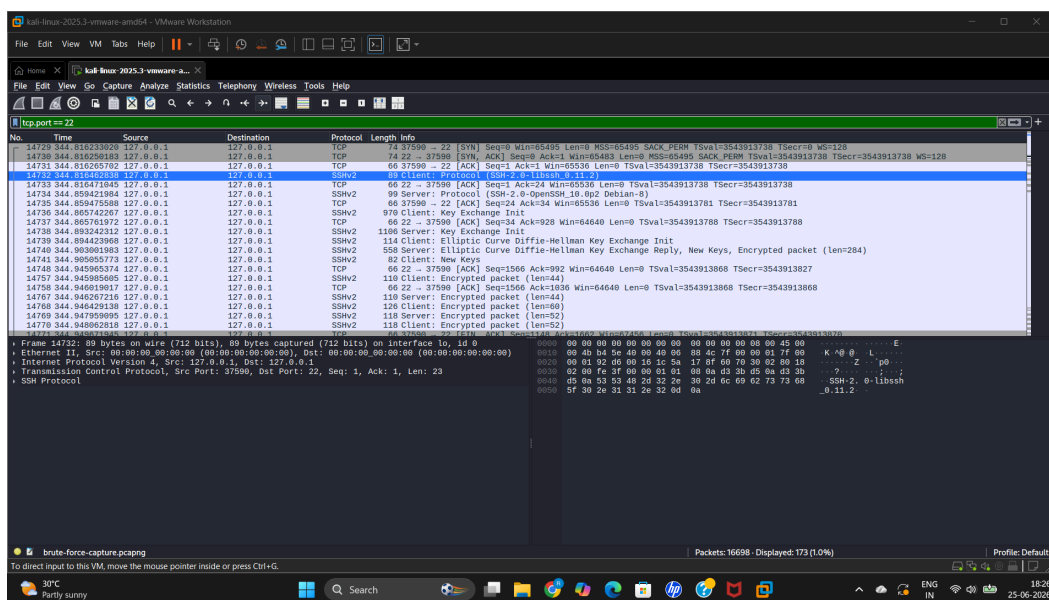
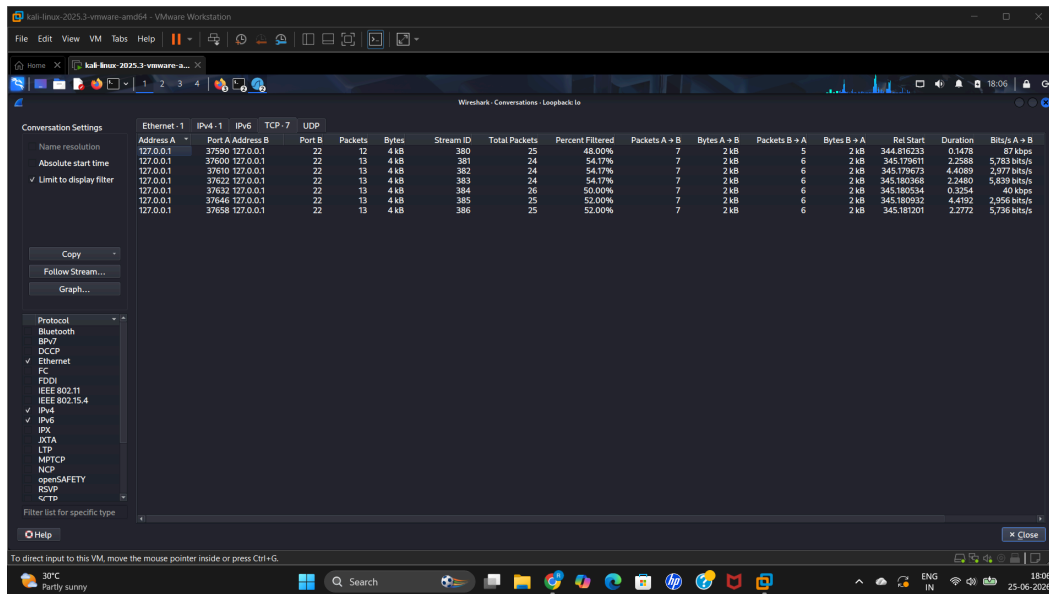
Interface: Loopback (lo)

Total Packets: 16,698

Filter Applied: tcp.port == 22 (173 packets displayed)

Key Findings:

- 7 TCP connections to port 22 (Stream IDs 380-386)
- Attack tool identified: SSH-2.0-libssh\_0.11.2 (Hydra fingerprint)
- Failed connections: RST packets — forceful disconnect
- Successful connection: FIN packet — graceful close
- IO Graph: Sudden traffic spike to 600+ packets/sec during attack
- Expert Information: Connection reset (RST) warnings on failed attempts



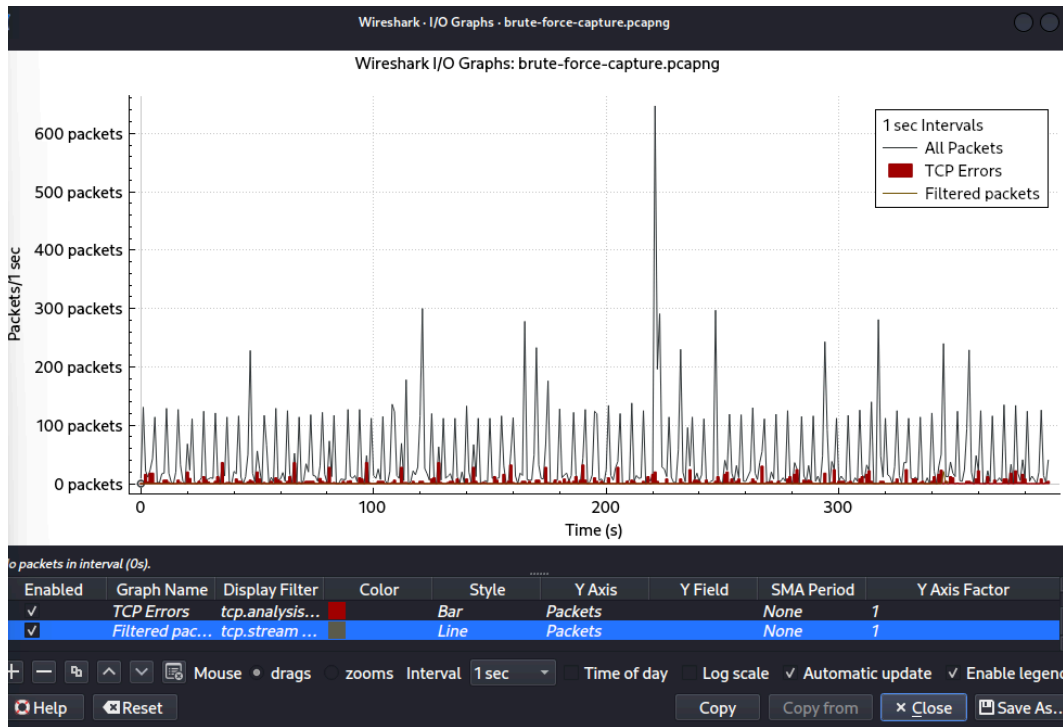


Figure 6: Wireshark — IO Graph (Traffic Spike)

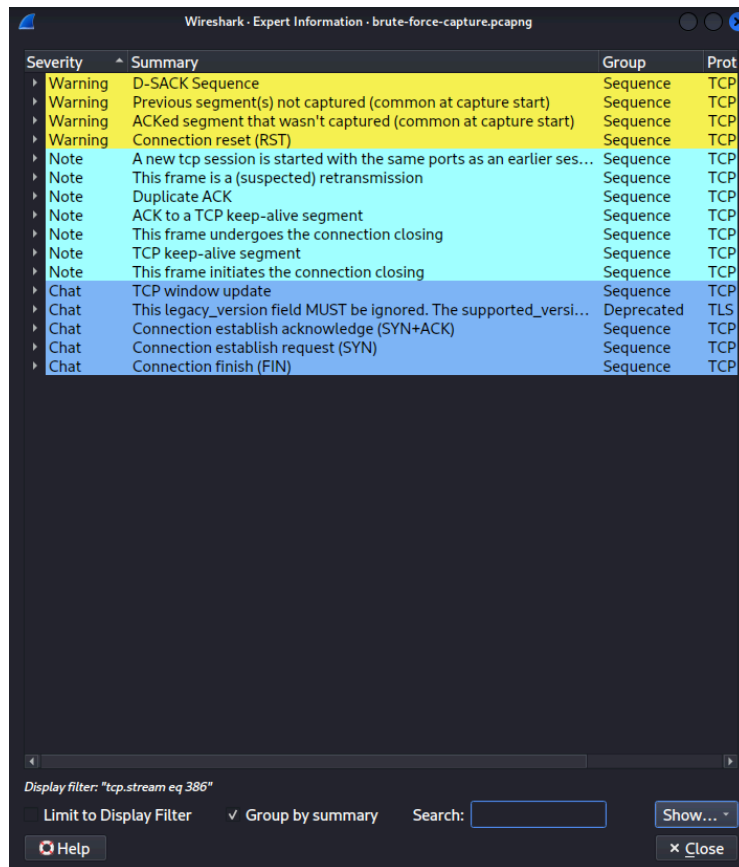


Figure 7: Wireshark — Expert Information (RST Packets)

## 5. MITRE ATT&CK MAPPING

Tactic: Credential Access

Technique: T1110 — Brute Force

Sub-technique: T1110.001 — Password Guessing

Detection: AN1275 — High volume failed logins followed by success

Attack Tool: Hydra v9.5

Target Protocol: SSH (Port 22)

Target Service: OpenSSH 10.0p2

Indicators of Compromise (IOCs):

- Source IP: 127.0.0.1 (lab) / External IP in real scenario
- User Agent: SSH-2.0-libssh\_0.11.2
- Port: 22
- Pattern: 5+ failed logins within 5 seconds followed by success

## 6. ROOT CAUSE ANALYSIS

Primary Cause:

Weak password policy — account password was same as username ("kali:kali")

This is a default credential commonly targeted in brute force attacks.

Contributing Factors:

- No account lockout policy configured
- No Multi-Factor Authentication (MFA) enabled
- SSH exposed without IP whitelisting
- No rate limiting on SSH authentication attempts

Detection Gap:

Attack occurred on June 25 but alert triggered on June 27 due to log ingestion delay in lab environment. In production, real-time log ingestion would trigger alert within 5 minutes of attack.

## 7. RECOMMENDATIONS

Immediate Actions:

1. Change default credentials — never use username as password
2. Enable account lockout — lock after 3 failed attempts for 30 minutes
3. Implement MFA — add second factor for SSH authentication

Short-term Actions:

4. Disable password authentication — use SSH key-based authentication only

5. Implement IP whitelisting — allow SSH only from trusted IPs
6. Enable fail2ban — automatically block IPs with multiple failed logins

Long-term Actions:

7. Deploy SIEM alerting in real-time — ensure log ingestion is immediate
8. Regular password audits — identify and remediate weak passwords
9. Network segmentation — limit SSH access to management network only
10. Regular penetration testing — identify weaknesses before attackers do

## **8. CONCLUSION**

This investigation confirms a brute force SSH attack where unauthorized access was achieved through password guessing using the Hydra tool. The attack was successful due to weak password policy and absence of account lockout mechanisms.

The incident was detected using Splunk SIEM with threshold-based alerting (5+ failed logins within 5 minutes) and confirmed through Wireshark network analysis. The attack tool was identified through SSH client fingerprinting (libssh\_0.11.2) in the PCAP capture.

Implementing the recommended controls — strong passwords, account lockout, MFA, and SSH key authentication — would have prevented this attack entirely.

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Date: June 27, 2026